

AI-Powered Sign Language Recognition and Gesture-Based Action Control

Real-time ASL alphabet recognition via webcam, powered by deep learning and deployed through a Streamlit application.

GROUP MEMBERS

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COURSE:

INTELLIGENT CONSUMER TECHNOLOGIES

Presentation Outline

- The problem we are trying to solve
- Analysis of our collected dataset
- Our proposed solution and system design
- Key assumptions behind the approach
- Model performance — effectiveness and efficiency
- Positive and negative results
- Operating conditions and practical use observations
- Conclusion and future work

The Problem We Are Solving

- Sign language is essential for Deaf and hard-of-hearing communities, yet everyday consumer technology rarely supports gesture-based interaction.
- Typing, searching, and opening applications typically require a keyboard or mouse — creating barriers for sign language users.
- Existing solutions often need specialized hardware or are limited to offline, non-interactive recognition.
- **Our challenge:** can a standard webcam and deep learning enable real-time ASL finger-spelling that controls real digital actions?

Why This Matters

- ASL finger-spelling is the foundation for communicating names, places, and unfamiliar words.
- Consumer technologies should be accessible without expensive or custom equipment.
- Combining recognition with action control moves beyond academic classification toward practical intelligent consumer applications.
- The project aligns with ICT goals: sensing, learning, and acting in a human-centered system.

Project Objectives

- Recognize static ASL alphabet signs (A–Y) from a live webcam feed in real time.
- Preprocess data, train, and compare a custom CNN with two transfer-learning models.
- Deploy an interactive Streamlit application for end users.
- Map recognized gestures to text generation and web-based actions.
- Evaluate effectiveness (accuracy) and efficiency (inference speed) under realistic conditions.

Dataset Overview

4,681

TOTAL IMAGES

26

LETTER FOLDERS

24

CLASSES USED

224²

INPUT SIZE

- Images organized in per-letter folders under dataset/.
- Each folder contains hand-sign photographs for one ASL letter.
- For training, **24 classes** are used (A–Y); J and Z are excluded (motion-based signs).
- Effective training set: **3,458 images** · Validation: **863 images** (80/20 split).

Data Characteristics

DISTRIBUTION

- Highly balanced: ~180 images per letter.
- Minor variation: Q and R have 178 images each.
- No severe class imbalance — suitable for multi-class classification.
- J and Z folders exist but are skipped at training time.

VISUAL PROPERTIES

- Static hand poses against varied backgrounds.
- RGB images resized uniformly to 224×224 px.
- At runtime, hands are converted to landmark skeletons on a white canvas.
- Representation shift from raw photos affects live performance.

The domain gap between dataset photos and live landmark drawings is a key factor in real-world performance.

Preprocessing & Augmentation

PREPROCESSING

- Train/validation split: 80% / 20%, seed = 42.
- Basic CNN & MobileNet: pixel rescaling ($\div 255$).
- EfficientNetB0: dedicated preprocess_input normalization.
- Batch size: 32 · Optimizer: Adam · Loss: categorical cross-entropy.

AUGMENTATION (TRANSFER LEARNING)

- Rotation $\pm 15^\circ$
- Width/height shift $\pm 10\%$
- Shear & zoom $\pm 10\%$
- Horizontal flip

Solution Overview

Webcam → Hand Detection → Landmark Canvas → CNN Classifier → Stability Filter → Action / Text

- **Input:** live webcam frame via OpenCV.
- **Detection:** cvzone HandDetector locates a single hand and 21 landmarks.
- **Classification:** one of three trained Keras models predicts the letter.
- **Output:** confirmed gesture triggers text append or opens a mapped web service.
- **Deployment:** Streamlit web app with configurable model, confidence, and mode.

Hand Detection & Feature Representation

- Hand bounding box extracted with 29 px padding around the detected region.
- 21 landmarks drawn as a skeleton on a 400 × 400 white canvas.
- Canvas resized to 224 × 224 and preprocessed per model requirements.
- Only **one hand** is processed per frame — multi-hand scenes are ignored.
- A **Stability filter** requires the same prediction across ~15 consecutive frames.
- Default **confidence threshold: 80%** · Action cooldown: **2.5 s**.

Model Architectures

Model	Type	Key Details	Epochs
Basic CNN	Custom	3 Conv2D blocks + Dense(256) + Dropout	10
MobileNetV2	Transfer learning	ImageNet weights, frozen base + custom head	15
EfficientNetB0	Transfer learning	ImageNet weights, model-specific preprocessing	15

MobileNetV2 is the default deployed model — balance of accuracy and lighter architecture.

Training & Evaluation Setup

TRAINING PROCESS

- Implemented in Jupyter notebooks under training/.
- Models saved as .keras files in models/.
- ~109 batches per epoch (batch size 32).
- ~2 min/epoch on CPU for transfer-learning models.

EVALUATION METRICS

- **Effectiveness:** validation accuracy & loss.
- **Efficiency:** FPS during live Streamlit inference.
- **Qualitative:** stability of predictions, usability of action modes.
- GPU not available on native Windows TensorFlow ≥ 2.11 — all benchmarks on CPU.

Streamlit Application

- Single-page app (app.py) with sidebar controls and live webcam preview.
- Users can switch between three models without restarting the session.
- Live metrics: current prediction, confidence, FPS, stable-frame counter.
- Event history logs the last 10 triggered actions with timestamps.
- ASL reference chart (signs.png) shown alongside the camera feed.
- Prototype desktop version also built with Tkinter (run_mediapipe.py).

Gesture-Based Action Mapping

TEXT GENERATION MODE

- Each confirmed letter is appended to a running text buffer.
- Users spell words letter by letter using ASL signs.
- Text can be cleared from the sidebar at any time.

WEB ACTIONS MODE

- Each letter maps to a web service (G → Google, Y → YouTube, E → Gmail).
- Sign S searches the generated text on Google.
- Actions open in the default browser; cooldown prevents duplicate triggers.

Text-to-speech (pyttsx3) is listed as a dependency for potential future integration.

Key Assumptions in Our Solution

SCOPE & ENVIRONMENT

- Only **static** ASL alphabet signs — not words, phrases, or motion signs (J, Z).
- Exactly **one hand** visible and reasonably centered in frame.
- Adequate, even lighting with an uncluttered background.
- Users perform signs similar to those in the training dataset.

TECHNICAL & INTERACTION

- Landmark skeletons generalize from dataset photos to live webcam input.
- A standard USB webcam provides sufficient image quality.
- Users will hold each sign steady for frame-stability confirmation.
- 80% confidence and 2.5 s cooldown are reasonable defaults for reliable triggering.

Effectiveness — Quantitative Results

Model	Train Acc.	Val. Accuracy	Val. Loss
Basic CNN	98.70%	99.30%	0.030
MobileNetV2	95.46%	98.49%	0.055
EfficientNetB0	94.42%	98.84% (peak)	0.065

- All models exceed **98% validation accuracy** on held-out data.
- Basic CNN achieves the highest final validation score.
- Stable-frame filtering and confidence threshold reduce false triggers in live use.

Efficiency — Runtime Performance

~5–10

AVG. FPS (STREAMLIT)

~2 min

PER TRAINING EPOCH

CPU

INFERENCE HARDWARE

- Benchmarked over 30–120 s sessions; logged in streamlit_fps_results.txt.
- 30 s run: avg. **4.96 FPS** · 60 s run: avg. **10.19 FPS**.
- FPS varies with model load, frame processing, and Streamlit rerun overhead.
- Not true real-time video (30 FPS) — acceptable for gesture-held input.
- GPU acceleration or model optimization could significantly improve throughput.

Under What Conditions It Works

WORKS WELL WHEN

- Lighting is even; background is plain and uncluttered.
- One hand is fully visible, centered, and held steady.
- Signs match training poses; confidence stays above 80%.
- User pauses between letters for stability confirmation.
- Web Actions mode used for single-letter shortcuts.

DEGRADES WHEN

- Multiple hands or partial occlusion appear in frame.
- Low light, motion blur, or fast hand transitions occur.
- Similar-looking signs (M / N / T) are performed quickly.
- J or Z motion signs are attempted.
- CPU is under heavy load — FPS drops below ~5.

Qualitative Observations & User Feedback

POSITIVE OBSERVATIONS

- Web Actions mode feels intuitive — gestures reliably open mapped services.
- Text Generation spells short words when signs are held deliberately.
- Adjustable confidence slider helps tune precision vs. responsiveness.
- Live landmark overlay gives clear feedback on hand detection.
- Model switching in the sidebar works seamlessly during a session.

NEGATIVE OBSERVATIONS

- Stability requirement (~15 frames) slows spelling — feels laggy.
- Similar signs occasionally confused despite high validation accuracy.
- Landmark representation differs from training photos.
- Low FPS on CPU makes the system feel less real-time than expected.
- No formal user study — feedback based on team testing sessions.

Summary & Future Work

- Built a complete pipeline: data → training → real-time recognition → gesture-driven actions.
- Three models compared; all achieved >98% validation accuracy on 24 ASL classes.
- Demonstrated practical consumer use cases beyond raw classification.
- Clearly defined assumptions, operating conditions, and known limitations.

- **FUTURE WORK**

Integrate text-to-speech · Add J/Z motion detection · GPU acceleration

- Train on landmark images to close the domain gap
- Expand to words and phrases · Conduct formal usability study

Thank you — Questions?

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